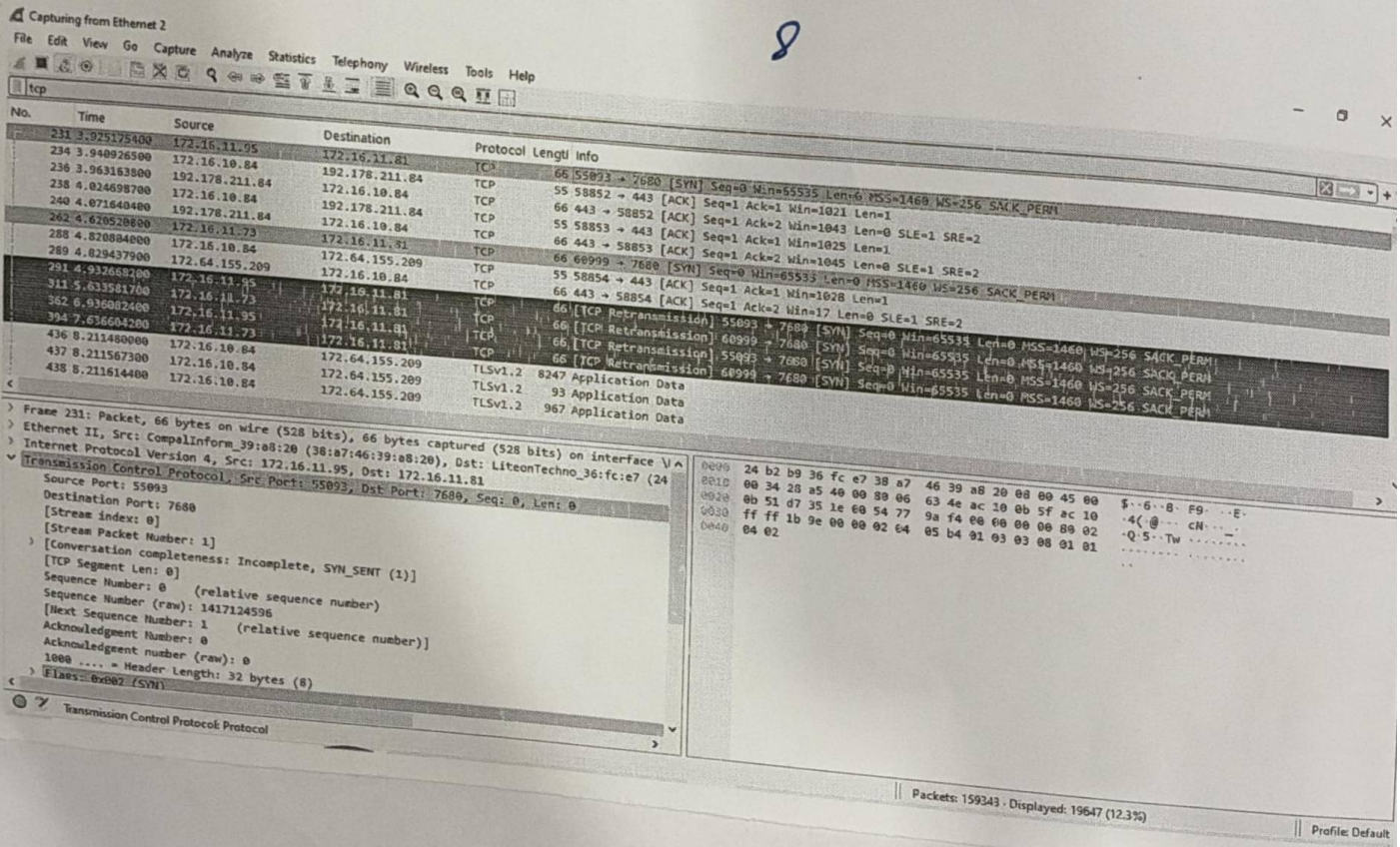
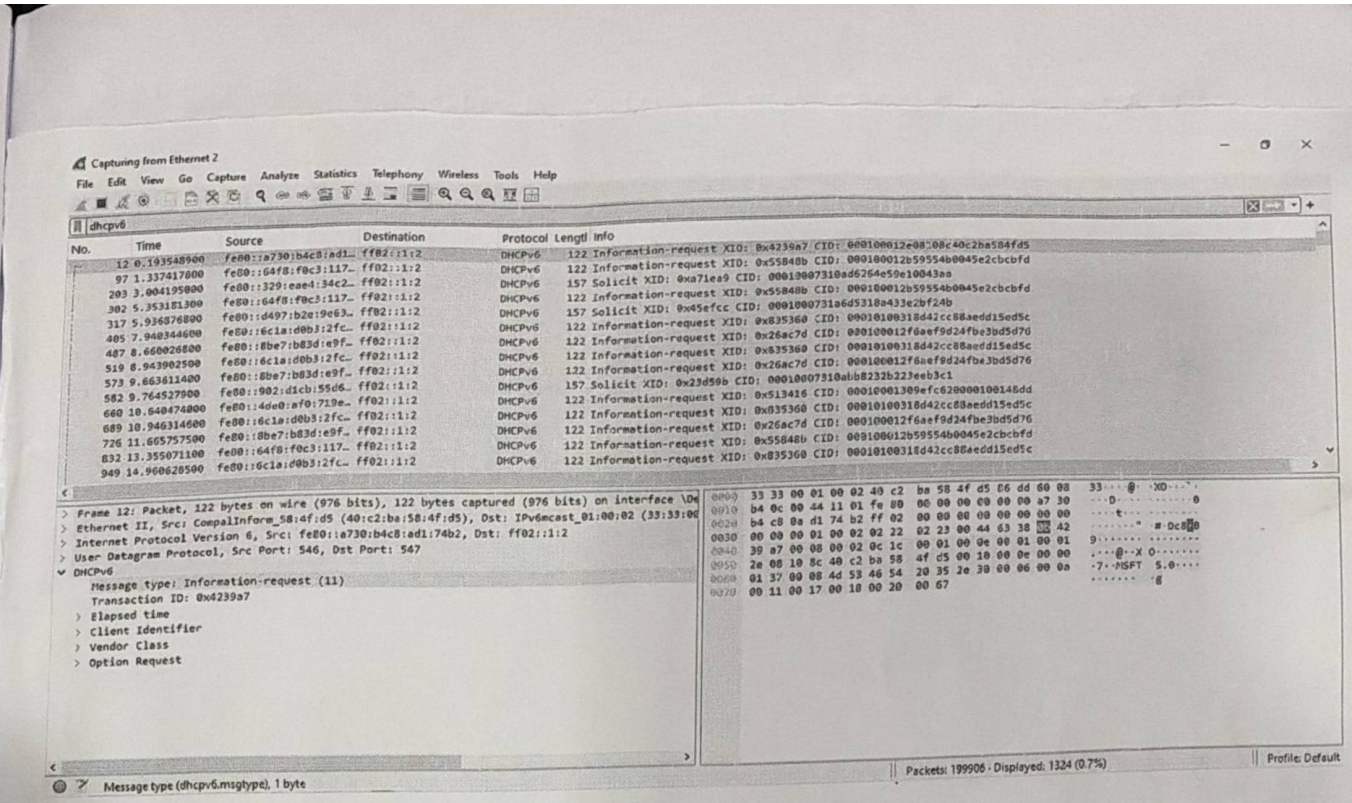
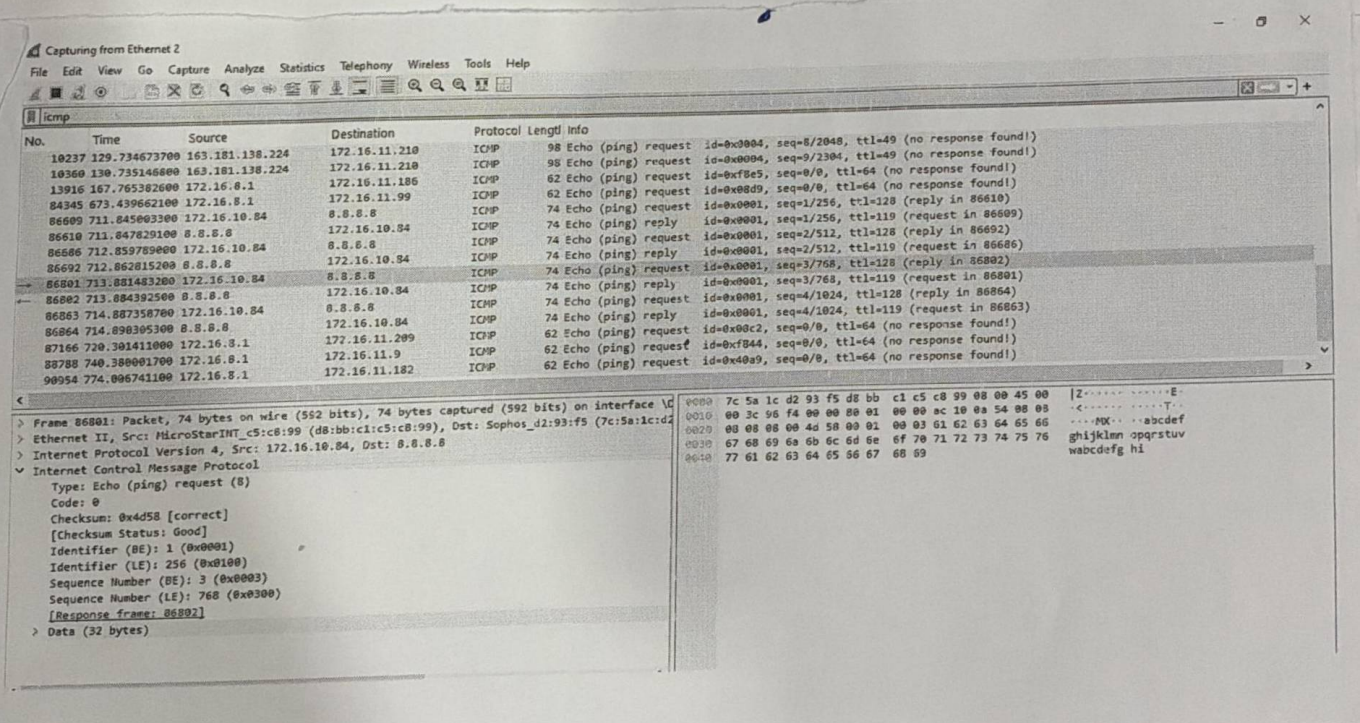
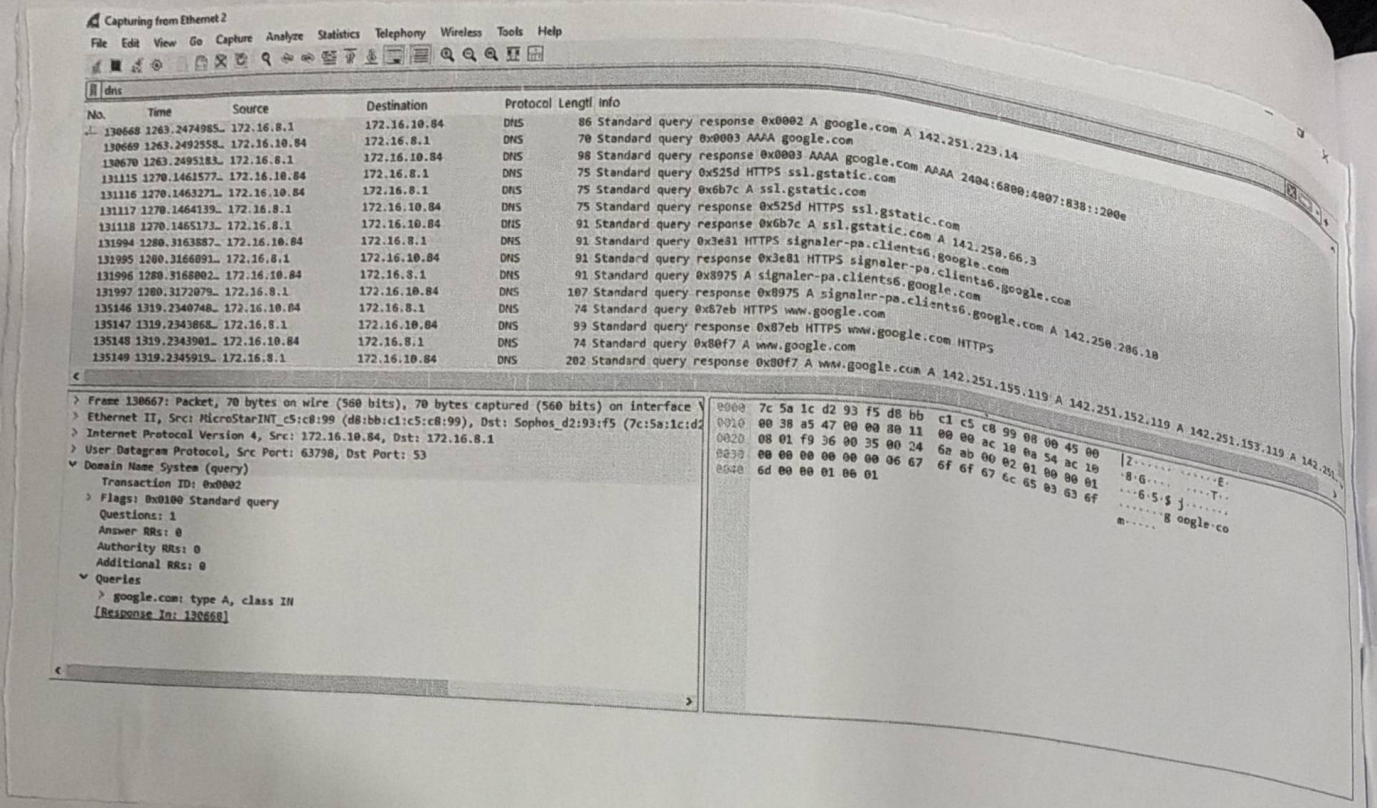
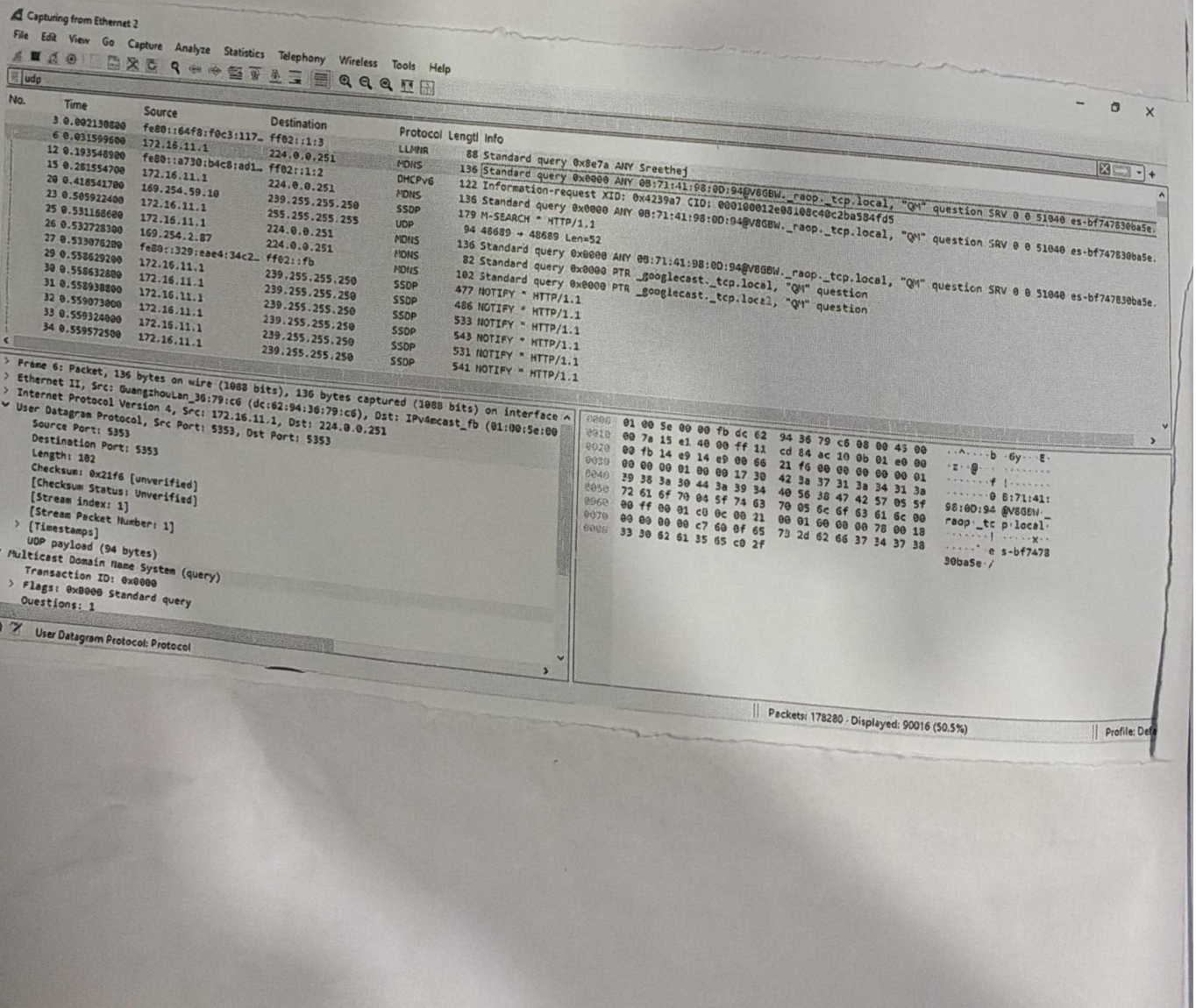
EXP NO : 8 TO CAPTURE SAVE AND ANALYSE NETWORK TRAFFIC ON TCP/UDP/IP/HTTP/ARP/DHCP/ICMP/ DNS using Wireshark Tool

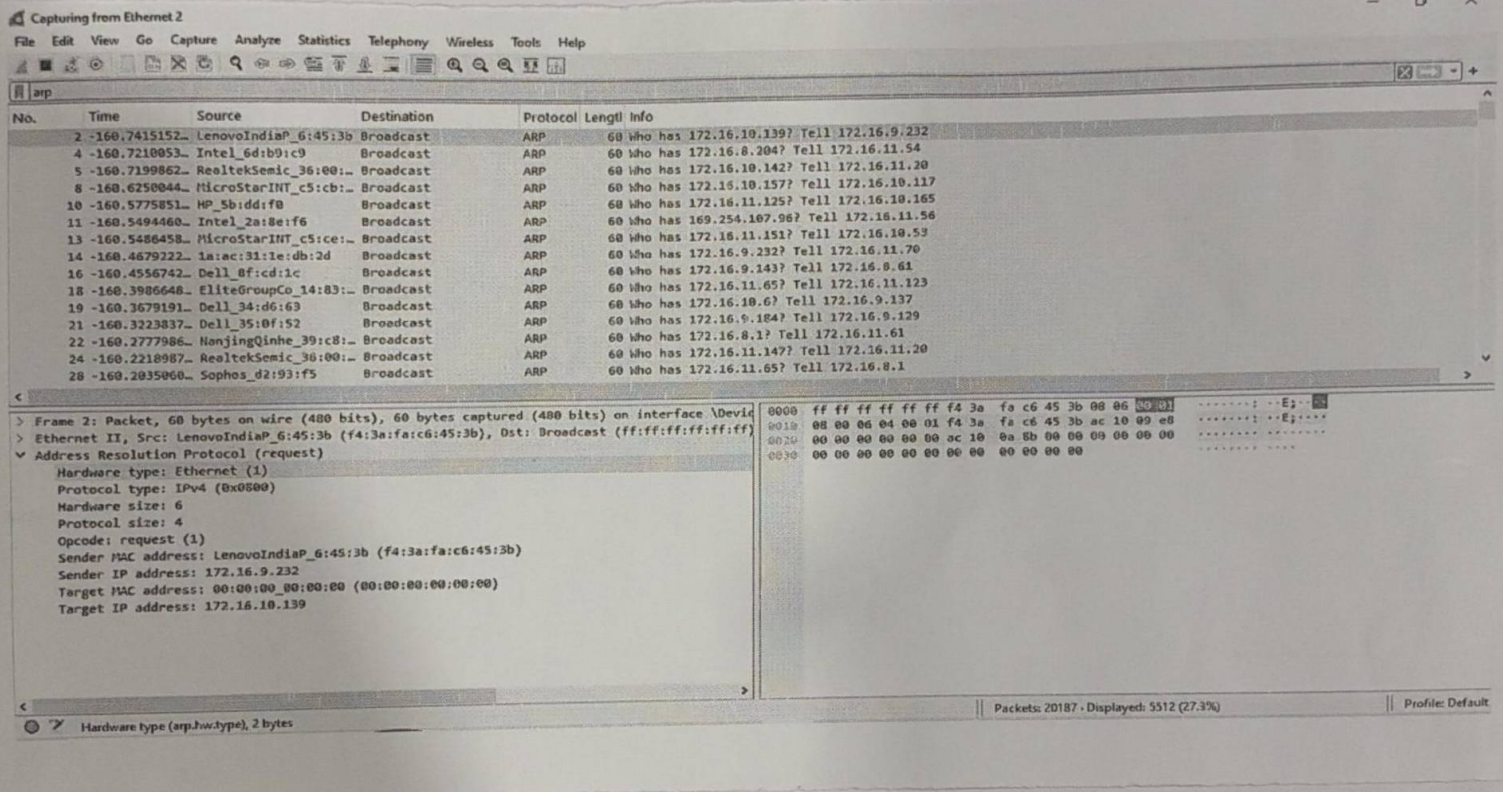



Capturing from Ethernet 2
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dhcpv6
No. Time Source Destination Protocol Length Info
12 0.193548900 fe80::a730:b4c8:ad1… ff02::1:2 DHCPv6 122 Information-request XID: 0x4239a7 CID: 000100012e08c08c40c2ba584fd5
97 1.337417600 fe80::64f8:f0c3:117… ff02::1:2 DHCPv6 122 Information-request XID: 0x55848b CID: 000100012b59554b0045e2cbcbfd
203 3.004195000 fe80::329:eae4:34c2… ff02::1:2 DHCPv6 157 Solicit XID: 0xa71ea9 CID: 00010007310ad6264e59e10043aa
302 5.353181300 fe80::64f8:f0c3:117… ff02::1:2 DHCPv6 122 Information-request XID: 0x55848b CID: 000100012b59554b0045e2cbcbfd
317 5.936876800 fe80::d497:b2e:9e63… ff02::1:2 DHCPv6 157 Solicit XID: 0x45efcc CID: 0001000731a6d5318a433e2bf24b
405 7.940344600 fe80::6c1a:d0b3:2fc… ff02::1:2 DHCPv6 122 Information-request XID: 0x835360 CID: 00010100318d42cc88aedd15ed5c
487 8.660026800 fe80::8be7:b83d:e9f… ff02::1:2 DHCPv6 122 Information-request XID: 0x26ac7d CID: 000100012f6aef9d24fbe3bd5d76
519 8.943902500 fe80::6c1a:d0b3:2fc… ff02::1:2 DHCPv6 122 Information-request XID: 0x835360 CID: 00010100318d42cc88aedd15ed5c
573 9.663611400 fe80::8be7:b83d:e9f… ff02::1:2 DHCPv6 122 Information-request XID: 0x26ac7d CID: 000100012f6aef9d24fbe3bd5d76
582 9.764527900 fe80::902:d1cb:55d6… ff02::1:2 DHCPv6 157 Solicit XID: 0x23d59b CID: 00010007310abb8232b223eeb3c1
660 10.640474000 fe80::4de0:af0:719e… ff02::1:2 DHCPv6 122 Information-request XID: 0x513416 CID: 00010001309efc620000100148dd
689 10.946314600 fe80::6c1a:d0b3:2fc… ff02::1:2 DHCPv6 122 Information-request XID: 0x835360 CID: 00010100318d42cc88aedd15ed5c
726 11.665757500 fe80::8be7:b83d:e9f… ff02::1:2 DHCPv6 122 Information-request XID: 0x26ac7d CID: 000100012f6aef9d24fbe3bd5d76
832 13.355071100 fe80::64f8:f0c3:117… ff02::1:2 DHCPv6 122 Information-request XID: 0x55848b CID: 000100012b59554b0045e2cbcbfd
949 14.960620500 fe80::6c1a:d0b3:2fc… ff02::1:2 DHCPv6 122 Information-request XID: 0x835360 CID: 00010100318d42cc88aedd15ed5c
Frame 12: Packet, 122 bytes on wire (976 bits), 122 bytes captured (976 bits) on interface \D
Ethernet II, Src: CompalInform_58:4f:d5 (40:c2:ba:58:4f:d5), Dst: IPv6mcast_01:00:02 (33:33:00
Internet Protocol Version 6, Src: fe80::a730:b4c8:ad1:74b2, Dst: ff02::1:2
User Datagram Protocol, Src Port: 546, Dst Port: 547
DHCPv6
Message type: Information-request (11)
Transaction ID: 0x4239a7
Elapsed time
Client Identifier
Vendor Class
Option Request
0000 33 33 00 01 00 02 40 c2 ba 58 4f d5 86 dd 60 08 33 @ XO `
0010 b4 0c 00 44 11 01 fe 80 00 00 00 00 00 00 a7 30 D 0
0020 b4 c8 0a d1 74 b2 ff 02 00 00 00 00 00 00 00 00 t
0030 00 00 00 01 00 02 02 22 02 23 00 44 63 38 DC 42 " # Dc8B
0040 39 a7 00 08 00 02 0c 1c 00 01 00 0e 00 01 00 01 9
0050 2e 08 10 8c 40 c2 ba 58 4f d5 00 10 00 0e 00 00 @ X O
0060 01 37 00 08 4d 53 46 54 20 35 2e 30 00 06 00 0a 7 MSFT 5.0
0070 00 11 00 17 00 18 00 20 00 67 g
Message type (dhcpv6.msgtype), 1 byte
Packets: 199906 · Displayed: 1324 (0.7%)
Profile: Default


Capturing from Ethernet 2
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
icmp
No. Time Source Destination Protocol Length Info
10237 129.734673700 163.181.138.224 172.16.11.210 ICMP 98 Echo (ping) request id=0x0004, seq=8/2048, ttl=49 (no response found!)
10360 130.735146800 163.181.138.224 172.16.11.210 ICMP 98 Echo (ping) request id=0x0004, seq=9/2304, ttl=49 (no response found!)
13916 167.765382600 172.16.8.1 172.16.11.186 ICMP 62 Echo (ping) request id=0xf8e5, seq=0/0, ttl=64 (no response found!)
84345 673.439662100 172.16.8.1 172.16.11.99 ICMP 62 Echo (ping) request id=0x08d9, seq=0/0, ttl=64 (no response found!)
86609 711.845003300 172.16.10.84 8.8.8.8 ICMP 74 Echo (ping) request id=0x0001, seq=1/256, ttl=128 (reply in 86610)
86610 711.847829100 8.8.8.8 172.16.10.84 ICMP 74 Echo (ping) reply id=0x0001, seq=1/256, ttl=119 (request in 86609)
86686 712.859789000 172.16.10.84 8.8.8.8 ICMP 74 Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 86692)
86692 712.862815200 8.8.8.8 172.16.10.84 ICMP 74 Echo (ping) reply id=0x0001, seq=2/512, ttl=119 (request in 86686)
86801 713.881483200 172.16.10.84 8.8.8.8 ICMP 74 Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply in 86802)
86802 713.884392500 8.8.8.8 172.16.10.84 ICMP 74 Echo (ping) reply id=0x0001, seq=3/768, ttl=119 (request in 86801)
86863 714.887358700 172.16.10.84 8.8.8.8 ICMP 74 Echo (ping) request id=0x0001, seq=4/1024, ttl=128 (reply in 86864)
86864 714.890305300 8.8.8.8 172.16.10.84 ICMP 74 Echo (ping) reply id=0x0001, seq=4/1024, ttl=119 (request in 86863)
87166 720.301411000 172.16.8.1 172.16.11.209 ICMP 62 Echo (ping) request id=0x00c2, seq=0/0, ttl=64 (no response found!)
88788 740.380001700 172.16.8.1 172.16.11.9 ICMP 62 Echo (ping) request id=0xf844, seq=0/0, ttl=64 (no response found!)
90954 774.006741100 172.16.8.1 172.16.11.182 ICMP 62 Echo (ping) request id=0x40a9, seq=0/0, ttl=64 (no response found!)
Frame 86801: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \D
Ethernet II, Src: MicroStarINT_c5:c8:99 (d8:bb:c1:c5:c8:99), Dst: Sophos_d2:93:f5 (7c:5a:1c:d2
Internet Protocol Version 4, Src: 172.16.10.84, Dst: 8.8.8.8
Internet Control Message Protocol
Type: Echo (ping) request (8)
Code: 0
Checksum: 0x4d58 [correct]
[Checksum Status: Good]
Identifier (BE): 1 (0x0001)
Identifier (LE): 256 (0x0100)
Sequence Number (BE): 3 (0x0003)
Sequence Number (LE): 768 (0x0300)
[Response frame: 86802]
Data (32 bytes)
0000 7c 5a 1c d2 93 f5 d8 bb c1 c5 c8 99 08 00 45 00 |Z E
0010 00 3c 96 f4 00 00 80 01 00 00 ac 10 0a 54 08 08 <
0020 08 08 08 00 4d 58 00 01 00 03 61 62 63 64 65 66 MX abcdef
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ghijklmn opqrstuv
0040 77 61 62 63 64 65 66 67 68 69 wabcdefg hi
Profile: Default


Capturing from Ethernet 2
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
No. Time Source Destination Protocol Length Info
130668 1263.2474985… 172.16.8.1 172.16.10.84 DNS 86 Standard query response 0x0002 A google.com A 142.251.223.14
130669 1263.2492558… 172.16.10.84 172.16.8.1 DNS 70 Standard query 0x0003 AAAA google.com
130670 1263.2495183… 172.16.8.1 172.16.10.84 DNS 98 Standard query response 0x0003 AAAA google.com AAAA 2404:6800:4807:838::200e
131115 1270.1461577… 172.16.10.84 172.16.8.1 DNS 75 Standard query 0x525d HTTPS ssl.gstatic.com
131116 1270.1463271… 172.16.8.1 172.16.10.84 DNS 75 Standard query 0x6b7c A ssl.gstatic.com
131117 1270.1464139… 172.16.10.84 172.16.8.1 DNS 75 Standard query response 0x525d HTTPS ssl.gstatic.com
131118 1270.1465173… 172.16.8.1 172.16.10.84 DNS 91 Standard query response 0x6b7c A ssl.gstatic.com A 142.250.66.3
131994 1280.3163887… 172.16.10.84 172.16.8.1 DNS 91 Standard query 0x3e81 HTTPS signaler-pa.clients6.google.com
131995 1280.3166891… 172.16.8.1 172.16.10.84 DNS 91 Standard query response 0x3e81 HTTPS signaler-pa.clients6.google.com
131996 1280.3168002… 172.16.10.84 172.16.8.1 DNS 91 Standard query 0x8975 A signaler-pa.clients6.google.com
131997 1280.3172079… 172.16.8.1 172.16.10.84 DNS 107 Standard query response 0x8975 A signaler-pa.clients6.google.com
135146 1319.2340748… 172.16.10.84 172.16.8.1 DNS 74 Standard query 0x87eb HTTPS www.google.com
135147 1319.2343868… 172.16.8.1 172.16.10.84 DNS 99 Standard query response 0x87eb HTTPS www.google.com A 142.250.206.10
135148 1319.2343901… 172.16.10.84 172.16.8.1 DNS 74 Standard query 0x80f7 A www.google.com HTTPS
135149 1319.2345919… 172.16.8.1 172.16.10.84 DNS 202 Standard query response 0x80f7 A www.google.com A 142.251.155.119 A 142.251.152.119 A 142.251.153.119 A 142.251
Frame 130667: Packet, 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface
Ethernet II, Src: MicroStarINT_c5:c8:99 (d8:bb:c1:c5:c8:99), Dst: Sophos_d2:93:f5 (7c:5a:1c:d2
Internet Protocol Version 4, Src: 172.16.10.84, Dst: 172.16.8.1
User Datagram Protocol, Src Port: 63798, Dst Port: 53
Domain Name System (query)
Transaction ID: 0x0002
Flags: 0x0100 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 0
Queries
google.com: type A, class IN
[Response In: 130668]
0000 7c 5a 1c d2 93 f5 d8 bb c1 c5 c8 99 08 00 45 00 |Z..... ..E.
0010 00 38 a5 47 00 00 80 11 00 00 ac 10 08 54 ac 10 ·8·G···· ·····T··
0020 08 01 f9 36 00 35 00 24 6a ab 00 02 01 00 00 01 ···6·5·$ j·····
0030 00 00 00 00 00 00 06 67 6f 6f 67 6c 65 03 63 6f ·····g oogle·co
0040 6d 00 00 01 00 01 m·····


Capturing from Ethernet 2
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
No. Time Source Destination Protocol Length Info
3 0.002130800 fe80::64f8:f0c3:117… ff02::1:3 LLMNR 88 Standard query 0x8e7a ANY Sreethej
6 0.031599600 172.16.11.1 224.0.0.251 MDNS 136 Standard query 0x0000 ANY 08:71:41:98:0D:94@V8GBW._raop._tcp.local, "QM" question SRV 0 0 51040 es-bf747830ba5e.
12 0.193548900 fe80::a730:b4c8:ad1… ff02::1:2 DHCPv6 122 Information-request XID: 0x4239a7 CID: 000100012e08108c40c2ba584fd5
15 0.281554700 172.16.11.1 224.0.0.251 MDNS 136 Standard query 0x0000 ANY 08:71:41:98:0D:94@V8GBW._raop._tcp.local, "QM" question SRV 0 0 51040 es-bf747830ba5e.
20 0.418541700 169.254.59.10 239.255.255.250 SSDP 179 M-SEARCH * HTTP/1.1
23 0.505922400 172.16.11.1 255.255.255.255 UDP 94 48689 → 48689 Len=52
25 0.531168600 172.16.11.1 224.0.0.251 MDNS 136 Standard query 0x0000 ANY 08:71:41:98:0D:94@V8GBW._raop._tcp.local, "QM" question SRV 0 0 51040 es-bf747830ba5e.
26 0.532728300 169.254.2.87 224.0.0.251 MDNS 82 Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
27 0.533076200 fe80::329:eae4:34c2… ff02::fb MDNS 102 Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
29 0.558629200 172.16.11.1 239.255.255.250 SSDP 477 NOTIFY * HTTP/1.1
30 0.558632800 172.16.11.1 239.255.255.250 SSDP 486 NOTIFY * HTTP/1.1
31 0.558938800 172.16.11.1 239.255.255.250 SSDP 533 NOTIFY * HTTP/1.1
32 0.559073000 172.16.11.1 239.255.255.250 SSDP 543 NOTIFY * HTTP/1.1
33 0.559324000 172.16.11.1 239.255.255.250 SSDP 531 NOTIFY * HTTP/1.1
34 0.559572500 172.16.11.1 239.255.255.250 SSDP 541 NOTIFY * HTTP/1.1
Frame 6: Packet, 136 bytes on wire (1088 bits), 136 bytes captured (1088 bits) on interface
Ethernet II, Src: GuangzhouLan_36:79:c6 (dc:62:94:36:79:c6), Dst: IPv4mcast_fb (01:00:5e:00
Internet Protocol Version 4, Src: 172.16.11.1, Dst: 224.0.0.251
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Source Port: 5353
Destination Port: 5353
Length: 102
Checksum: 0x21f6 [unverified]
[Checksum Status: Unverified]
[Stream index: 1]
[Stream Packet Number: 1]
[Timestamps]
UDP payload (94 bytes)
Multicast Domain Name System (query)
Transaction ID: 0x0000
Flags: 0x0000 Standard query
Questions: 1
User Datagram Protocol: Protocol
0000 01 00 5e 00 00 fb dc 62 94 36 79 c6 08 00 45 00 ··^····b ·6y···E·
0010 00 7a 15 e1 40 00 ff 11 cd 84 ac 10 0b 01 e0 00 ·z··@··· ····
0020 00 fb 14 e9 14 e9 00 66 21 f6 00 00 00 00 00 01 ·····f !·····
0030 00 00 00 01 00 00 17 30 42 3a 37 31 3a 34 31 3a ·····0 B:71:41:
0040 39 38 3a 30 44 3a 39 34 40 56 38 47 42 57 05 5f 98:0D:94 @V8GBW._
0050 72 61 6f 70 04 5f 74 63 70 05 6c 6f 63 61 6c 00 raop._tc p·local·
0060 00 ff 00 01 c0 0c 00 21 00 01 00 00 00 78 00 18 ·····! ·····x··
0070 00 00 00 00 c7 60 0f 65 73 2d 62 66 37 34 37 38 ·····` ·es-bf7478
0080 33 30 62 61 35 65 c0 2f 30ba5e·/
Packets: 178280 · Displayed: 90016 (50.5%)
Profile: Def


Capturing from Ethernet 2
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
arp
No. Time Source Destination Protocol Lengtl Info
2 -160.7415152… LenovoIndiaP_6:45:3b Broadcast ARP 60 Who has 172.16.10.139? Tell 172.16.9.232
4 -160.7210053… Intel_6d:b9:c9 Broadcast ARP 60 Who has 172.16.8.204? Tell 172.16.11.54
5 -160.7199862… RealtekSemic_36:00:… Broadcast ARP 60 Who has 172.16.10.142? Tell 172.16.11.20
8 -160.6250044… MicroStarINT_c5:cb:… Broadcast ARP 60 Who has 172.16.10.157? Tell 172.16.10.117
10 -160.5775851… HP_5b:dd:f0 Broadcast ARP 60 Who has 172.16.11.125? Tell 172.16.10.165
11 -160.5494460… Intel_2a:8e:f6 Broadcast ARP 60 Who has 169.254.107.96? Tell 172.16.11.56
13 -160.5486458… MicroStarINT_c5:ce:… Broadcast ARP 60 Who has 172.16.11.151? Tell 172.16.10.53
14 -160.4679222… 1a:ac:31:1e:db:2d Broadcast ARP 60 Who has 172.16.9.232? Tell 172.16.11.70
16 -160.4556742… Dell_8f:cd:1c Broadcast ARP 60 Who has 172.16.9.143? Tell 172.16.8.61
18 -160.3986648… EliteGroupCo_14:83:… Broadcast ARP 60 Who has 172.16.11.65? Tell 172.16.11.123
19 -160.3679191… Dell_34:d6:63 Broadcast ARP 60 Who has 172.16.10.6? Tell 172.16.9.137
21 -160.3223837… Dell_35:0f:52 Broadcast ARP 60 Who has 172.16.9.184? Tell 172.16.9.129
22 -160.2777986… NanjingQinhe_39:c8:… Broadcast ARP 60 Who has 172.16.8.1? Tell 172.16.11.61
24 -160.2218987… RealtekSemic_36:00:… Broadcast ARP 60 Who has 172.16.11.147? Tell 172.16.11.20
28 -160.2035060… Sophos_d2:93:f5 Broadcast ARP 60 Who has 172.16.11.65? Tell 172.16.8.1
> Frame 2: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Devic
> Ethernet II, Src: LenovoIndiaP_6:45:3b (f4:3a:fa:c6:45:3b), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Address Resolution Protocol (request)
Hardware type: Ethernet (1)
Protocol type: IPv4 (0x0800)
Hardware size: 6
Protocol size: 4
Opcode: request (1)
Sender MAC address: LenovoIndiaP_6:45:3b (f4:3a:fa:c6:45:3b)
Sender IP address: 172.16.9.232
Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
Target IP address: 172.16.10.139
0000 ff ff ff ff ff ff f4 3a fa c6 45 3b 08 06 00 01
0010 08 00 06 04 00 01 f4 3a fa c6 45 3b ac 10 09 e8
0020 00 00 00 00 00 00 ac 10 0a 8b 00 00 00 00 00 00
0030 00 00 00 00 00 00 00 00 00 00 00 00
Hardware type (arp.hw.type), 2 bytes
Packets: 20187 · Displayed: 5512 (27.3%)
Profile: Default